

Assignment 1

30 Marks

Deadline: Jul 1, 2026 (Wednesday)

Question - 1:

Can you relate/connect Amdahl's law with any of the design principles, mentioned below? Explain the reason with an example. [5]

	Design Principle
1	Performance via Prediction
2	Performance via Pipelining
3	Make the common case faster
4	Use Abstraction to Simplify Design

Question - 2:

Define the term throughput and response time. How do these two differ, and why might one be prioritized over the other in different computing environments? [5]

Question - 3:

Consider three different processors P1, P2, and P3 executing the exact same instruction set. P1 has a 4 GHz clock rate and a CPI of 2. P2 has a 2.5 GHz clock rate and a CPI of 1.5. P3 has a 5.0 GHz clock rate and has a CPI of 2.7.

Now answer the following questions:

- Which processor has the highest performance expressed in instructions per second?
- We are trying to reduce the execution time by 40%, but this leads to an increase of 35% in the CPI. What clock rate should we have to get this time reduction?

[3 + 2]

Question - 4:

Two computers, A and B, execute the same set of instructions.

- Computer A: Clock Rate = 3 GHz, CPI = 1.5
- Computer B: Clock Rate = 2.5 GHz, CPI = 1.2

Which one is faster? Show calculations for CPU time. [5]

Question - 5:

A processor has the following instruction mix:

- 50% ALU instructions with CPI = 1
- 30% load/store with CPI = 2
- 20% branch with CPI = 3

Calculate the **average CPI** of the program. [5]

Question - 6:

A computer runs a program in 20 seconds. A compiler optimization reduces instruction count by 15% and CPI by 10%, but the clock rate stays the same. What is the new execution time? [5]